

04-21-2025

50/50 new + old material on final
↳ focus on "big topics"

Event Study:

- event time, e

↳ combo of group and time period

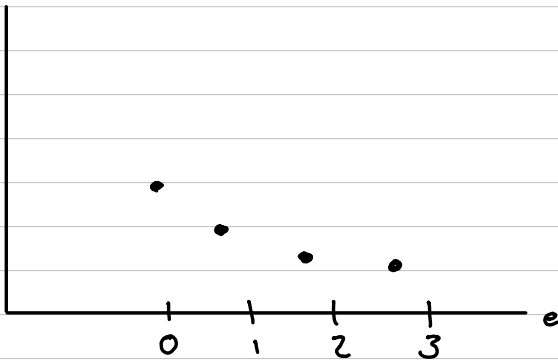
- ex: $e=0$, $T=10$, $G_i=5$

$$t = G_i + 0 = 5$$

↳ $e=1$, $T=10$, $G_i=5$

$$t = G_i + 1 = 6$$

- plots



$\mathcal{U}_e$ - set of groups observed to participate in treatment for e periods.

$$\hookrightarrow \mathcal{U}_0 = \mathcal{U} \setminus \{\infty\}$$

$$\hookrightarrow \mathcal{U}_1 = \mathcal{U} \setminus \{\infty, T\}$$

Event Study:

for units that satisfy $G_i \in \mathcal{G}_c$, define

$$TE_i(c) = Y_{i, G_i+c} - Y_{i, G_i+c}(\infty)$$

$$ATT^c(c) = E[TE(c) | G \in \mathcal{G}_c]$$

$$(LIE) = \sum_{g \in \mathcal{G}_c} P(G=g | G \in \mathcal{G}_c) ATT(g, g+c)$$

↳ weighted average of group-time ATE
s.t. bigger groups are weighted more

$$ATT(g, t) = E[Y_t - Y_t(\infty) | G=g]$$

Panel Data Fixed Effects Regressions:

$$Y_{it} = \theta + \eta_i + \alpha D_{it} + e_{it}$$

- assuming T.E. homogeneity can consistently estimate model

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$$Y_{it} = \theta + \eta_i + \tilde{X}_{it}^T \tilde{\beta} + e_{it} \quad (1)$$

s.t.  $\tilde{X}_{it}^T = (1 \{t=1\}, \dots, 1 \{t=T\}, \tilde{x}_{it})$

$$\bar{Y}_i = \frac{1}{T} \sum_{t=1}^T Y_{it}; \quad \bar{X}_i = \frac{1}{T} \sum_{t=1}^T \tilde{X}_{it}; \quad \bar{e}_i = \frac{1}{T} \sum_{t=1}^T e_{it}$$

Assume strict exogeneity: for  $t=1, \dots, T$

$$E[e_{it} | \underline{X}_i] = 0 \quad \text{s.t. } \underline{X}_i \text{ is } t \times k = [X_{i1}^T, \dots, X_{iT}^T]^T$$

## Panel Data FE:

- Strict exog: model is linear, rule out error term effecting  $x$ 's
- within transformation:  $\dot{C}_{it} = C_{it} - \bar{C}_i$
- $\therefore \dot{Y}_{it} = Y_{it} - \bar{Y}_i$

$$\bar{Y}_i = \bar{X}_i^T \beta + \eta_i + \bar{e}_i \quad (2)$$

$$(1) - (2): Y_{it} - \bar{Y}_i = (X_{it}^T - \bar{X}_i^T) \beta + \eta_i - \eta_i + e_{it} - \bar{e}_i$$

$$\dot{Y}_{it} = \dot{X}_{it}^T \beta + \dot{e}_{it}$$

notation:

$\underline{1}$ :  $T \times 1$  vector of 1's

$\underline{Y}_i$ :  $T \times 1$  vector of  $Y_{it}$  for  $t = 1, \dots, T$

$$\underline{\dot{Y}}_i: \underline{Y}_i - \underline{1} \bar{Y}_i \quad // \quad \bar{Y}_i = (\underline{1}^T \underline{1})^{-1} \underline{1}^T \underline{Y}_i$$

$$\underline{\dot{Y}}_i = \underline{Y}_i - \underbrace{\underline{1} (\underline{1}^T \underline{1})^{-1} \underline{1}^T}_{\text{annihilator}} \underline{Y}_i$$

$$= \underline{m} \underline{Y}_i \quad \text{s.t. } \underline{m} \text{ is } T \times T$$

by same arguments

$$\underline{\dot{X}}_i = \underline{m} \underline{X}_i \quad \text{s.t. } \underline{m} \text{ is } T \times k$$

- aside:

$$\hat{\beta} = \left( \frac{1}{n} \sum_{i=1}^n X_i X_i^T \right)^{-1} \frac{1}{n} \sum_{i=1}^n X_i Y_i$$

- now:

$$\hat{\beta} = \left( \frac{1}{nT} \sum_{i=1}^n \sum_{t=1}^T \dot{X}_i \dot{X}_i^T \right)^{-1} \frac{1}{nT} \sum_{i=1}^n \sum_{t=1}^T \dot{X}_i \dot{Y}_i$$

↳ can cancel  $\frac{1}{nT}$ 's.

$$= \left( \sum_{i=1}^n \underline{\dot{X}}_i^T \underline{\dot{X}}_i \right)^{-1} \sum_{i=1}^n \underline{\dot{X}}_i^T \underline{\dot{Y}}_i$$

$$= \left( \sum_{i=1}^n \underline{\dot{X}}_i^T \underline{m} \underline{\dot{X}}_i \right)^{-1} \sum_{i=1}^n \underline{\dot{X}}_i^T \underline{m} \underline{\dot{Y}}_i$$

↳  $\underline{m}$  is symm. and idemp.

↳ check notes for comment on estimation

$$\underline{Y}_i = \underline{X}_i \beta + \underline{\varepsilon}_i$$

$$\Rightarrow \hat{\beta} - \beta = \left( \sum_{i=1}^n \underline{X}_i^T \underline{m} \underline{X}_i \right)^{-1} \sum_{i=1}^n \underline{X}_i \underline{m} \underline{\varepsilon}_i$$

1: unbiasedness:  $\underline{X} : n \times k$

$$E[\hat{\beta} - \beta | \underline{X}] = \left( \sum_{i=1}^n \underline{X}_i^T \underline{m} \underline{X}_i \right)^{-1} \sum_{i=1}^n \underline{X}_i \underline{m} E[\underline{\varepsilon}_i | \underline{X}]$$

by strict exog:  $E[\underline{\varepsilon}_i | \underline{X}] = 0 \Rightarrow E[\hat{\beta} - \beta] = 0$   
 $\Rightarrow \hat{\beta}$  unbiased for  $\beta$